

Project Workflow: Weeks 8–10

Phase: Scoring, Polish, Testing & Final Submission

1. Objective

The goal of Weeks 8 through 10 was to complete the gameplay loop by introducing progressive scoring, a real-time HUD, item management, and win/loss states, followed by rigorous bug fixing, system performance verification, and project documentation. This included:

- Implementing a comprehensive scoring system tracking car passes and gas collection.
- Building a live HUD overlay displaying score, speed, and distance progress.
- Designing functional Game Over and You Won screens with an R-key restart feature.
- Conducting exhaustive testing on collision detection, lane switching, road curves, and maintaining a smooth 60 FPS performance.
- Preparing final project documentation and completing the submission.

2. Detailed Task Breakdown

- **Implementing Scoring, HUD, and Gas Items:** Developed a real-time scoring logic rewarding players +10 points per car passed and +5 points per gas canister collected. Integrated a heads-up display (HUD) overlay rendering live score, current speed, and distance progress. Designed state transition screens for "Game Over" and "You Won" alongside a keyboard listener enabling instant game restart via the R key.
- **Bug Fixing & Comprehensive Testing:** Performed rigorous testing across core gameplay mechanics, specifically validating collision boundaries, lane-switching responsiveness, and road curve behavior. Addressed and resolved bugs related to boost mechanics, lane transition edge cases, and win/loss trigger conditions. Verified that the rendering engine maintained a consistent 60 FPS gameplay experience without frame drops.
- **Final Testing, Documentation & Submission:** Executed complete end-to-end system testing across all edge cases and gameplay scenarios. Compiled all required

project documentation, technical implementation summaries, and game screenshots, culminating in the final project packaging and submission.

3. Deliverables

- **Scoring & Item Mechanics:** Fully functional point system (+10 for cars, +5 for gas items) and interactive fuel collection.
- **HUD Overlay:** Live on-screen display tracking score, speed metrics, and distance progress.
- **State Screens:** Polished Game Over and You Won screens supporting quick restarts via the R key.
- **Testing & Performance Reports:** Verified bug-free mechanics (lane switching, collision, curves) optimized for stable 60 FPS execution.
- **Final Documentation:** Complete project report package, system screenshots, and final source code submission.

4. Tools & Resources

- **Development Environment:** Java JDK 8+, IntelliJ IDEA / VS Code.
- **Reference Materials:** Java 2D graphics rendering loops, game-state management patterns, keyboard event handling for restarts, and frame-rate optimization strategies for 60 FPS performance.